

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. STB-401: Sample Surveys and Design of Experiments

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. All questions carry equal marks. Symbols have their usual meanings.*

Question 1 (Compulsory).

[14]

- (a) The households in a town are to be sampled in order to estimate the average amount of assets per household. The households are stratified into a high-rent stratum and a low-rent stratum. A household in the high-rent stratum is thought to have about nine times as much assets as one in the low-rent stratum, and the stratum standard deviation S_h is expected to be proportional to the square root of the stratum mean. There are 4,000 households in the high-rent stratum and 20,000 in the low-rent stratum. Distribute a sample of 1,000 households between the two strata using optimum allocation. [7]
- (b) In a stratification with two strata, $W_1 = 0.8$, $W_2 = 0.2$, $S_1 = 2$ and $S_2 = 4$. Compute the sample sizes n_1 and n_2 (ignoring the finite population correction) if the standard error of the estimated population mean $\bar{y}_{st}$ is to be 0.1 and the total sample size $n = n_1 + n_2$ is to be minimized. [7]

Question 2.

- (a) Discuss in detail the different types of errors involved in sample surveys. [7]
- (b) Consider a population of 6 units with values 1, 2, 3, 4, 5, 6. List all possible samples of size 2 (without replacement) from this population. Verify that the sample mean is an unbiased estimator of the population mean. Also calculate the sampling variance of the sample mean and verify that:
- (i) it agrees with the formula for the variance of the sample mean under SRSWOR, and
- (ii) this variance is less than the variance of the sample mean obtained under SRSWR. [7]

Question 3.

- (a) Discuss the process of determination of sample size for obtaining an estimate with a specified coefficient of variation. [7]
- (b) Discuss ratio, regression, and product methods of estimation. Mention their relative importance in sampling theory. [7]

Question 4.

- (a) Describe the Analysis of Variance (ANOVA) for the fixed-effects model for one-way classified data. Also discuss the use and method of obtaining the critical difference (CD) in this context. [7]
 - (b) Mention the advantages and disadvantages of the Completely Randomised Design (CRD), Randomised Block Design (RBD), and Latin Square Design (LSD). [7]
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Question 5.

- (a) Explain the procedure of estimating one missing observation in an RBD. [7]
 - (b) What are the basic principles of design of experiments? Explain each with a suitable example. Also discuss the layout of a Latin Square Design. [7]
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Question 6.

- (a) Describe stratified random sampling. Write its advantages over simple random sampling and give the formula for estimating the population mean. [7]
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Question 7.

What are the advantages of:

- (a) simple random sampling over systematic sampling?
- (b) LSD over RBD?
- (c) objective sampling over subjective sampling?
- (d) sampling over complete enumeration?

[14]

Question 8. Write short notes on any **two** of the following:

[14]

- (a) Systematic sampling
 - (b) Factorial experiments
 - (c) F -test and its applications
 - (d) Cluster sampling and two-stage sampling
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